

GOVARTHAN S

Junior AI Engineer | Machine Learning Engineer | Computer Vision Engineer | Data Scientist

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PROFESSIONAL SUMMARY

Results-driven ECE graduate with ISRO-deployed real-time telemetry and high-availability systems experience, uniquely bridging aerospace-grade reliability engineering with modern AI/ML development. Built mission-critical dual-redundant data acquisition infrastructure at ISRO Propulsion Complex (IPRC), Mahendragiri, monitoring live propulsion engine tests with fault-tolerant failover logic, multithreaded concurrent processing, and zero data loss. Proficient in Python ML pipelines (Scikit-Learn, XGBoost, Pandas, NumPy), computer vision (OpenCV, YOLOv8, DeepFace), and Generative AI engineering (LangChain, FAISS, Transformers, LLMs, RAG, multi-agent systems via CrewAI and LangGraph). Targeting AI/ML engineering roles in industrial intelligence, computer vision, and enterprise Generative AI domains.

TECHNICAL SKILLS

Languages	Python SQL MATLAB
Machine Learning	Regression Classification Clustering Feature Engineering Model Evaluation XGBoost Scikit-Learn
Deep Learning	ANN CNN Transfer Learning PyTorch TensorFlow Keras
Computer Vision	OpenCV YOLOv8 DeepFace Object Detection Image Processing
Generative AI	LLMs RAG Prompt Engineering Agentic AI LangChain LangGraph CrewAI Hugging Face
Data Science	EDA Data Cleaning Statistical Analysis Visualization Pandas NumPy Matplotlib Seaborn
Databases	MySQL PostgreSQL SQLite FAISS (Vector DB)
Tools & DevOps	Git GitHub Docker FastAPI Streamlit Jupyter Power BI Tableau
Cloud	AWS (Basics) Google Cloud Platform (Basics)
Real-Time Systems	Multithreading UDP Socket Programming Queue Buffering Failover Architecture DewesoftX PXI
Embedded	Raspberry Pi Sensor Interfacing GPIO

WORK EXPERIENCE

ISRO Propulsion Complex (IPRC), Mahendragiri, Tamil Nadu

Jan 2026 – Mar 2026

Project Trainee — Mission-Critical DAQ & Real-Time Telemetry Systems

- Engineered production-deployed dual-redundant real-time telemetry monitoring system at India's premier space agency, eliminating single points of failure in aerospace data acquisition infrastructure during live propulsion engine test campaigns at IPRC Mahendragiri.
- Implemented dual-threaded UDP packet receiver architecture sustaining 50ms continuous monitoring cycles across multi-channel propulsion sensor streams, achieving zero data loss under simulated server failure conditions.
- Designed watchdog-based server health monitoring with automatic primary-to-redundant failover switching and configurable cooldown hysteresis, preventing oscillatory behavior in fault-tolerant mission-critical infrastructure.
- Developed queue-based circular buffer for real-time telemetry sample management and non-blocking Matplotlib visualization engine sustaining continuous waveform rendering across 6+ weeks of propulsion engine testing.
- Integrated CDT timestamp synchronization module enabling precise temporal mapping of dual-channel acquisition data for accurate propulsion event sequencing and control-room telemetry correlation.

PROJECTS

AI-Powered Smart Surveillance & Threat Detection System

[Python](#) | [OpenCV](#) | [YOLOv8](#) | [DeepFace](#) | [FastAPI](#) | [Streamlit](#)

- Built real-time multi-camera surveillance system integrating YOLOv8 object detection and DeepFace facial recognition for simultaneous threat classification and person identification at 30 FPS with sub-100ms inference latency per frame.
- Engineered FastAPI REST backend for camera management and alert subscriptions, and Streamlit dashboard for live feed monitoring, configurable detection zones, and forensic event logging.

Predictive Maintenance System for Industrial Equipment

[Python](#) | [Scikit-Learn](#) | [XGBoost](#) | [Power BI](#) | [SQL](#)

- Developed end-to-end predictive maintenance ML pipeline on multi-sensor equipment data (vibration, temperature, pressure) using XGBoost, achieving 94% failure prediction accuracy with 48-hour advance warning; applied SMOTE to improve minority-class recall from 61% to 89%.
- Built automated feature engineering module (35+ time-series features) with SQL data pipeline and interactive Power BI dashboard visualizing equipment health scores, failure probability trends, and maintenance scheduling.

AI Resume Screening & ATS Analyzer

[Python](#) | [LangChain](#) | [FAISS](#) | [Transformers](#) | [Streamlit](#)

- Engineered RAG-based intelligent resume screening system using LangChain, FAISS vector search, and Hugging Face sentence-transformers for semantic JD-resume matching, LLM-powered skill gap analysis, and automated ATS score estimation beyond keyword overlap.
- Built Streamlit multi-page application supporting bulk resume upload, automated candidate ranking across multiple JDs, skill gap visualization, and PDF report export for HR workflow automation.

Multi-Agent AI Business Assistant

[CrewAI](#) | [LangGraph](#) | [LLMs](#) | [FastAPI](#)

- Architected multi-agent AI business automation system using CrewAI and LangGraph with specialized Researcher, Analyst, Writer, and QA agents; LangGraph manages stateful workflow orchestration, conditional routing, and error recovery across complex multi-step task pipelines.
- Integrated tool-augmented agents (web search, SQL query, code execution) with FastAPI service layer exposing async task execution, status polling, and structured JSON output for enterprise business intelligence automation.

Manufacturing Analytics Dashboard

[Power BI](#) | [Python](#) | [SQL](#) | [PostgreSQL](#)

- Built enterprise manufacturing analytics dashboard in Power BI connected to PostgreSQL backend, visualizing OEE, production throughput, defect rates, and equipment utilization; engineered Python ETL pipeline and optimized SQL views, reducing dashboard query time by 65%.

Additional Projects:

IoT Industrial Monitoring & Redundant Switching System (Python, UDP, Matplotlib) — Primary-redundant server architecture with socket timeout failover and real-time visualization. | **Satellite Environmental Monitoring System** (Raspberry Pi, Python) — Multi-sensor data acquisition and real-time visualization. | **GlowGrid Smart Lighting** (Bluetooth, PWM, Circuit Design) — Audio-synchronized LED system with wireless control.

EDUCATION

B.E. Electronics & Communication Engineering 2022 – 2026
Government College of Engineering, Dharmapuri | **CGPA: 7.2** | Class XII: 85.6% | Class X: 96.6%

CERTIFICATIONS (IN PROGRESS / PLANNED)

Machine Learning Specialization — DeepLearning.AI | Deep Learning Specialization — DeepLearning.AI | IBM Data Science Professional Certificate | Google Data Analytics Professional Certificate | Microsoft Power BI Data Analyst | AWS Cloud Practitioner | Generative AI with LLMs — DeepLearning.AI